MedEvolve: Longitudinal Medical Task Generation through Dual Exploration

Anonymous ACL submission

Abstract

Longitudinal medical agent tasks require selecting clinical relationships that a patient record can support and that available tools can retrieve—not merely facts that appear in a tool trace. We present MedEvolve, which couples *data exploration* of supportable event patterns with *environment exploration* of evidence retrieval, then generates questions, answers, and source-level claims conditioned on both guides and verifies them against original records. Across PhysicianBench and MedAgentBench, TaskCraft verifies at 88–98% yet yields essentially no higher-order tasks among verified outputs, whereas MedEvolve verifies fewer candidates but produces substantially more integrative and decision-level ones: a depth–reliability tradeoff. Feedback-driven strategy updates across patient batches show mixed held-out effects and do not guarantee improvement.

1 Introduction and Related Work

MedAgentBench and PhysicianBench place medical agents in interactive EHR environments where they must retrieve records and integrate information to complete clinical tasks (Jiang et al., 2025; Liu et al., 2026). Building such tasks is not only selecting queryable facts. When questions span encounters, answers rest on relationships among treatments, tests, and changes over time. Which relationships are supportable varies by patient: serial measurements around a treatment change may support response analysis, whereas an isolated encounter may support only a local judgment.

Tool traces describe how an agent accesses information, not whether the retrieved facts form a coherent clinical question. Locating medication and laboratory records does not establish that measurements are comparable around a treatment change; a recorded subsequent action is not automatically the unique correct plan. Longer trajectories therefore do not by themselves imply deeper reasoning, and solver failures can stem from unsupported or ambiguous tasks.

Prior generators address parts of this problem. TaskCraft builds atomic tasks around extracted answers and expands retrieval and composition dependencies (Shi et al., 2025), but longitudinal EHRs additionally require a prior judgment of which relationships the current case can support. CuES and AgentEvolver discover tasks through environment interaction and assess feasibility via execution guidance or reference solutions (Mai et al., 2025; Zhai et al., 2025); successful retrieval or replay still does not alone validate clinical interpretation across patients. Environment-only guidance (as in our EnvExplore baseline) likewise improves tool awareness without deciding which clinical relations a given trajectory supports.

MedEvolve closes this gap with dual exploration: data exploration identifies supportable clinical relationships and evidence conditions; environment exploration identifies how to retrieve that evidence with available tools. Shared verification against original records filters unsupported candidates, and optional solver feedback revises generation guidance across patient batches while the solver remains fixed. We evaluate valid yield, quality, and reasoning depth in two medical environments, treating strategy updates as a secondary analysis rather than a guaranteed gain.

2 Method

2.1 Tasks and Strategies

Let X_p be the longitudinal trajectory of patient p. A task is $\tau = (p, q, a, E)$ with question q, candidate reference answer a, and evidence E as source identifiers with factual claims. The solver receives the question and patient context and accesses records through environment tools; reference answers and evidence claims remain on the evaluation side.

Data and environment exploration produce fixed

guides K_D and K_E. At round r, the strategy is $\theta_r = (g_r, W_r)$: natural-language generation guidance g_r and weighted directions $W_r = \{(c_j, d_j, w_j)\}_j$ over clinical categories (diagnosis/interpretation, testing/risk stratification, treatment planning, prescribing, and replanning) and difficulty descriptions. Each generation slot samples a direction proportional to w_j. The sampled direction constrains category and reasoning demand; the patient's evidence decides whether a concrete task is supportable.

2.2 Dual Exploration, Generation, and Verification

Data exploration summarizes event roles, temporal relations, and applicability conditions from evolution patients into K_D (e.g., comparing measurements around a treatment anchor requires comparable pre/post records). Environment exploration uses read-only tools to record accessible record types and retrieval procedures as K_E. The guides connect clinical evidence requirements with retrieval capability (Figure 1) and remain fixed after the initial exploration; subsequent feedback updates only g_r and W_r. Interface details are in Appendix A.

For patient p, one model call receives the budgeted trajectory, both guides, current guidance, and two sampled directions $Z_{p,r}$, returning at most two candidates:

$$\mathcal{T}_{p,r} = G(X_p, K_D, K_E, g_r, Z_{p,r}), \qquad |\mathcal{T}_{p,r}| \le 2.$$

The generator selects an event relationship, then forms the question, candidate answer, and source-level claims. Shared verification rereads cited records, checks each source–claim pair, and tests whether the claims jointly support the question and answer. Verified tasks proceed to quality and reasoning-depth assessment and independent solving; rejected tasks contribute defect reasons to feedback.

2.3 Feedback-Driven Updates

A feedback model aggregates per-patient verification outcomes, quality/difficulty judgments, and solver traces into F_r. An update model revises the strategy:

$$\theta_{r+1} = U_{\mathrm{LLM}}(\theta_r, F_r).$$

Guidance accumulates revisions to event selection, evidence organization, and answer formulation; weights adjust subsequent sampling. Patient batches are disjoint across rounds. Held-out evaluation freezes the strategy, fixes the initial direction list and sampling seed, and tests updated generation guidance rather than the full evolved direction distribution. An example revision is shown in Appendix A.3.

3 Experimental Setup

We evaluate on PhysicianBench and MedAgentBench with generation pools of 108 and 100 patients. Using seed 42, 27/25 patients are reserved for evolution and 81/75 for held-out evaluation; patients do not cross sets, and held-out outcomes do not enter updates (Appendix B). Reported runs use two evolution rounds. All methods generate at most two retrospective read-only tasks per patient and share held-out patients and external evaluation.

We report GPT-4o-mini as the primary generation/solver setting in Table 1 and summarize GPT-5.4-nano in text. Baselines: TaskCraft (atomic generation with depth/width expansion) (Shi et al., 2025); Workflow (localize–generate–filter); SingleShot (one-shot generation); SelfRefine (within-batch self-feedback) (Madaan et al., 2023); EnvExplore (environment guidance with a fixed initial strategy). MedEvolve uses dual exploration and cross-patient updates. Method budgets and implementation notes are in Appendix C.

Verification and completion judgments use GPT-5.4-nano. Quality and difficulty are scored independently by GPT-5.4-nano and GPT-4o-mini; we take the floor of the median of two valid scores. Quality covers factual accuracy (FA), clinical validity (CV), evidence sufficiency (ES), and question clarity (QC) on a 1–5 scale. Reasoning levels are factual retrieval, single-step reasoning, integrative reasoning, and clinical decision-making; levels 3–4 count as higher-order (HO). Verification pass rate uses tasks with verification results as the denominator; quality means use verified tasks with valid scores (Appendix D).

4 Results

4.1 Valid Yield, Quality, and Reasoning Depth

Table 1 reports generation yield, verification, quality, and higher-order counts for GPT-4o-mini on held-out patients (at most two tasks per patient; generation failures remain in N_G).

The central finding is a depth–reliability tradeoff. TaskCraft verifies at 88.2–94.4% under GPT-4o-mini, yet contributes *zero* higher-order verified

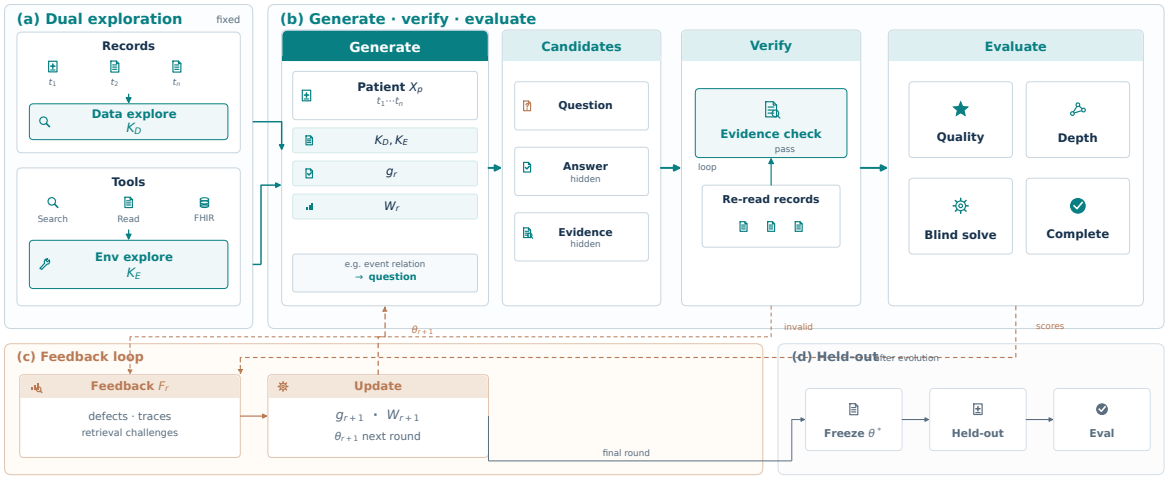


Figure 1: Overview of MedEvolve. (a) Dual exploration yields fixed data guidance K_D and environment guidance K_E. (b) Left-to-right: generate candidates from patient X_p, guides, g_r, and weights W_r; verify evidence against source records; evaluate quality, depth, blind solving, and completion. (c) Defects and traces form feedback F_r that updates g_{r+1}, W_{r+1} (dashed loop). (d) After evolution, freeze θ^* and evaluate on held-out patients.

Table 1: Main generation comparison with GPT-4o-mini on held-out patients. N_G: generated tasks; N_V: verified tasks; $V\%$: pass rate among tasks with verification results; FA/CV/ES/QC: mean quality (1–5) on verified scored tasks; HO: level 3–4 tasks / tasks with difficulty scores. PB: PhysicianBench; MAB: MedAgentBench.

Env	Method	N_G	N_V	$V\%$	FA	CV	ES	QC	HO
PB	MedEvolve	161	32	19.9	4.16	3.84	3.62	4.56	16/32
	TaskCraft	119	105	88.2	4.93	4.50	4.89	4.91	0/105
	Workflow	139	25	18.0	4.08	3.84	3.76	4.76	15/25
	SingleShot	162	86	53.1	4.42	4.23	4.14	4.87	10/86
	SelfRefine	101	26	25.7	3.96	4.00	3.58	4.38	12/26
	EnvExplore	160	27	16.9	4.04	3.96	3.52	4.74	14/27
MAB	MedEvolve	150	30	20.0	4.17	3.90	3.63	4.83	21/30
	TaskCraft	108	102	94.4	4.91	4.40	4.93	4.90	0/102
	Workflow	103	23	22.3	3.87	3.87	3.57	4.57	18/23
	SingleShot	150	76	50.7	4.39	4.16	4.08	4.70	16/76
	SelfRefine	111	19	17.1	3.84	3.84	3.53	4.74	9/19
	EnvExplore	141	26	18.4	4.04	3.92	3.65	4.65	16/26

tasks: outputs are predominantly factual retrieval. MedEvolve verifies at about 20%, but half or more of its verified tasks are higher-order (16/32 on PhysicianBench; 21/30 on MedAgentBench). That lower pass rate reflects a strict source–claim reread filter rather than post-hoc cherry-picking: HO share must be read with N_V, not in isolation. EnvExplore and Workflow similarly concentrate higher-order tasks in a smaller verified set, while SingleShot sits between TaskCraft and MedEvolve on both yield and HO count. High post-filter quality can accompany low yield. Appendix F gives a validity breakdown (Figure 3).

Under GPT-5.4-nano the same qualitative contrast holds against TaskCraft: TaskCraft verifies at 97.7% (PhysicianBench; HO 0/125) and 97.3% (MedAgentBench; HO 0/108), while MedEvolve verifies at 44.6% (HO 52/69) and 71.2% (HO 90/101). EnvExplore also yields many higher-order verified tasks under this stronger model (86/107 and 95/108), so dual exploration is not uniquely responsible for HO volume once the generator is strong; the sharper contrast remains MedEvolve vs. TaskCraft on the depth–reliability tradeoff. SelfRefine produces only 7–12 tasks in these runs, so its pass rates and means should not be overinterpreted.

4.2 Strategy Updates: Mixed Held-Out Effects

Comparing the initial strategy with the strategy after successive updates on a fixed held-out patient set (exploration guides, models, initial direction allocation, seed, and evaluation fixed; held-out

scores excluded from updates) shows no consistent improvement. Under GPT-4o-mini, PhysicianBench verification rises from 13.8% to 18.5% while HO verified count moves from 16 to 14; MedAgentBench verification falls from 11.0% to 6.7% (HO 12→10). Under GPT-5.4-nano, MedAgentBench verification falls from 80.7% to 71.7% (HO 95→78), whereas PhysicianBench verification is roughly flat (72.4%→73.8%; HO 93→99). Strategy self-evolution is therefore a secondary finding: updates can revise questioning conditions, but do not guarantee better held-out yield or higher-order verified volume (Figures 4–7).

5 Conclusion

MedEvolve generates longitudinal medical agent tasks by coupling data exploration of supportable clinical relationships with environment exploration of evidence retrieval, then verifying source-level claims against original records. Across two EHR environments, high verification yield and higher-order reasoning among verified tasks trade off: TaskCraft is reliable but shallow; MedEvolve is less reliable but deeper among accepted tasks. Feedback-driven strategy updates have mixed held-out effects and should not be assumed to improve generation.

Limitations

Record completeness, temporal information, and model context budgets limit discoverable questioning patterns; trajectory truncation may omit essential evidence. Patient-disjoint splits assess transfer only within an environment. Generation uses retrospective access without temporal-view isolation and therefore does not represent information available at a historical decision point.

Automated evaluation has shared-model bias: aggregating two LLM judges does not remove correlated errors, and existing records may not suffice to judge every clinical plan. Quality scores cover only verified tasks with valid scoring outputs; a complex task may be rejected because a single citation fails verification. Means must be read with coverage and valid yield.

Held-out evaluation fixes direction allocation and primarily tests updated generation guidance, not the full evolved direction distribution. Feedback diagnoses may be incorrect, and strategy updates need not improve held-out metrics monotonically. Component ablations and downstream experience-learning comparisons are not reported in this short version; Appendix E records the experience-learning protocol for completeness. Harness-specific learning triggers further limit cross-harness interpretation of any future downstream results.[1]

[1] Organization and evidence-status checks were assisted by the scientific-writing skill from Scientific Agent Skills (Kassis et al., 2026); authors remain responsible for literature and scientific claims.

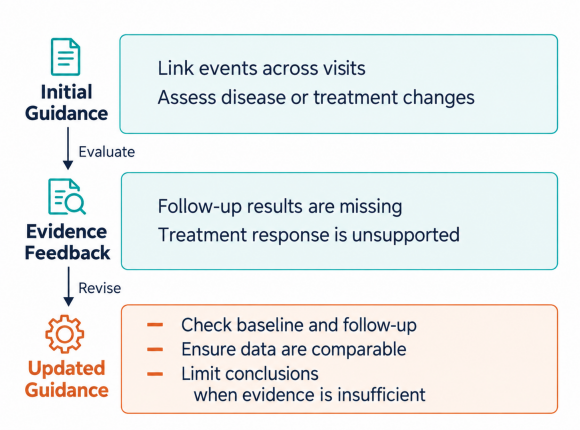


Figure 2: Example feedback-driven revision of generation guidance g_r. Verification feedback flags insufficient follow-up evidence; the revised guidance adds baseline/follow-up and comparability checks and restricts conclusions when evidence is weak. The example illustrates how g_r can change, not that updates improve held-out metrics. Text is paraphrased from one evolution trace.

A Exploration and Generation Interfaces

A.1 Data Exploration

Exploration tools expose open patient IDs, budgeted trajectories, and optional compressed summaries. Inputs retain patient identifiers, source pointers, and coverage flags; the output is patient-agnostic natural-language guidance K_D. The guide summarizes data organization, temporal change, event relations, evidence citation patterns, and supportable questioning conditions, without retaining individual clinical values or source IDs.

Compressed summaries locate candidate patterns; original records support concrete claims. Missing records and budget truncation are stated separately. Time windows, record counts, and clinical relations in K_D are inductive questioning constraints, not hard-coded medical rules applied uniformly across patients.

A.2 Environment Exploration

Environment exploration queries native read-only tools in an isolated instance over open patients. Guidance K_E records validated parameters, return shapes, pagination and truncation behavior, and typical retrieval procedures, and distinguishes call failures from unverified capabilities. Patient-specific observations stay out of the shared guide.

Table 2: Generation-side patient splits (seed 42).

Env	Pool	Evo.	Held-out	Rounds
PhysicianBench	108	27	81	13/14
MedAgentBench	100	25	75	12/13

A.3 Guidance Revision Example

A.4 Task Generation and Strategies

A task records patient ID, clinical category, question, candidate reference answer, source identifiers with per-source factual claims, and execution requirements. The shared generation interface leaves the temporal-view field empty and requires read-only execution. One generation call receives the budgeted trajectory, K_D, K_E, current guidance g_r, and two sampled directions $Z_{p,r}$, and returns at most two candidates.

The strategy is $\theta_r = (g_r, W_r)$ with natural-language guidance and weighted directions over five clinical categories and difficulty descriptions. Directions are sampled proportional to weights (at least one positive). K_D and K_E stay fixed during updates; only g_r and W_r change. Evolution rounds sample from current weights; held-out evaluation freezes the initial direction list and sampling seed while reading the updated g_r.

B Environments, Patients, and Information Access

Table 2 freezes evolution and held-out lists under seed 42 with prior exploration-exposure constraints; patients do not cross sets. Trajectories are ordered by record time and budgeted for model input; verification rereads original sources rather than input summaries. Initial exploration and generation expose query tools only. The public interface does not enable temporal-view isolation, so time language in a question does not hide later records server-side. Experience-learning train/aux splits over synthetic tasks are separate from this generation split and are not mixed with it.

C Generation Methods and Budgets

TaskCraft maps patient records and read-only FHIR tools onto atomic extraction, verification, and depth/width expansion; both branches start from atomic tasks, share a generation budget, and record acceptance separately from final filtering. Workflow runs localize–generate–filter with filtering only. SingleShot emits at most two tasks in one

pass. SelfRefine drafts, self-critiques, and revises within a patient batch (up to three critiques and two revisions; early stop on pass). EnvExplore obtains environment guidance once, then generates under a fixed initial strategy. MedEvolve uses dual exploration; reported runs also apply two rounds of cross-patient Feedback-Driven Updates, treated as a secondary analysis with mixed held-out effects rather than a guaranteed gain. All methods share the output schema and external evaluation; internal calls and filters are logged separately.

Matching task counts are not treated as identical token budgets. Exploration and updates count toward MedEvolve cost; shared evaluation is reported separately; reused historical calls are deduplicated from original logs.

D Scoring, Aggregation, and Statistics

D.1 Quality and Reasoning Depth

Quality uses four 1–5 dimensions scored independently: factual accuracy (FA), clinical validity (CV), evidence sufficiency (ES), and question clarity (QC). Rough anchors: 5 = fully accurate/supported/clear; 3 = mostly correct with material gaps; 1 = severe error, unsupported inference, or unusable question. Reasoning depth is labeled from task demand and evidence structure: (1) factual retrieval, (2) single-step judgment, (3) cross-record integration, (4) clinical decision-making; levels 3–4 count as higher-order (HO). Category and depth are recorded separately; treatment-planning items are not auto-labeled as level 4.

For task i and dimension d, two judges yield $s_{i,d}^{(1)}, s_{i,d}^{(2)}$; we take

$$s_{i,d} = \left\lfloor \operatorname{median}\left(s_{i,d}^{(1)}, s_{i,d}^{(2)}\right) \right\rfloor .$$

Depth uses the same aggregation. The protocol requires two valid judge outputs; a single success does not impute a missing score.

D.2 Evidence Structure and Rate Denominators

Deduplicated cited sources S_i give evidence count $|S_i|$; resource-type count uses recognizable type prefixes. Parseable timestamps T_i define span $\Delta_i = (\max T_i - \min T_i)$ in days (missing if none; zero if one). Span summaries use non-missing values only.

Zero denominators are reported as missing. Means and distributions are accompanied by coverage counts.

Table 3: Primary rate denominators.

Metric	Numerator	Denominator
Verif. coverage	w/ verification	generated
Verif. pass rate	verified	w/ verification
HO share	depth 3–4	w/ depth scores
Dim. HQ share	score ≥ 4	scored on dim.

Table 4: Harness experience carriers (protocol only; no scores).

Harness	Persisted experience	Test mode
OpenClaw	skills + state dir	frozen skills; learning off
Hermes	skills + memory	read-only; writes off
ACE	playbook	Generator only
Reflexion	transferable tips	Actor only; frozen tips

D.3 Uncertainty Estimation

Unless noted, intervals use patient-cluster bootstrap: 1,000 resamples, seed 42, 2.5/97.5 percentiles. All tasks of a patient enter together; method contrasts on shared patients stay paired. Zero-denominator resample draws stay missing rather than zero.

E Experience-Learning Protocol

Downstream experience learning is out of scope for reported results in this short paper; we record only the frozen protocol. Each harness starts from a shared initial state with isolated configs and experience stores. Training uses a fresh session and medical instance per task; answers and evidence are available only after execution for review/learning. After training, experience is frozen; test runs disable writes and background learning and start each task from the frozen snapshot.

Reflexion tips store applicability, steps, and failure lessons without patient IDs, individual values, or per-task answers (at most three retrieved tips under frozen rules). Official evaluation assets and native scorers, when used, are version-locked and never replaced by the shared LLM judge.

F Additional Results

Main-text Table 1 summarizes GPT-4o-mini held-out yield and quality; HO rates are conditional on the verified set size N_V. Figure 3 shows verification status and reasoning depth for the primary PhysicianBench / GPT-4o-mini setting. Figures 4–7 show descriptive held-out curves under Feedback-Driven Updates (fixed patient set, initial direction

allocation, and sampling seed); effects are mixed across settings and do not imply that evolution always helps. Missing checkpoints are not connected; error bars are patient-cluster 95% intervals when available.

G Illustrative Cases

Author-verified illustrative cases are pending and are not included in this draft.—

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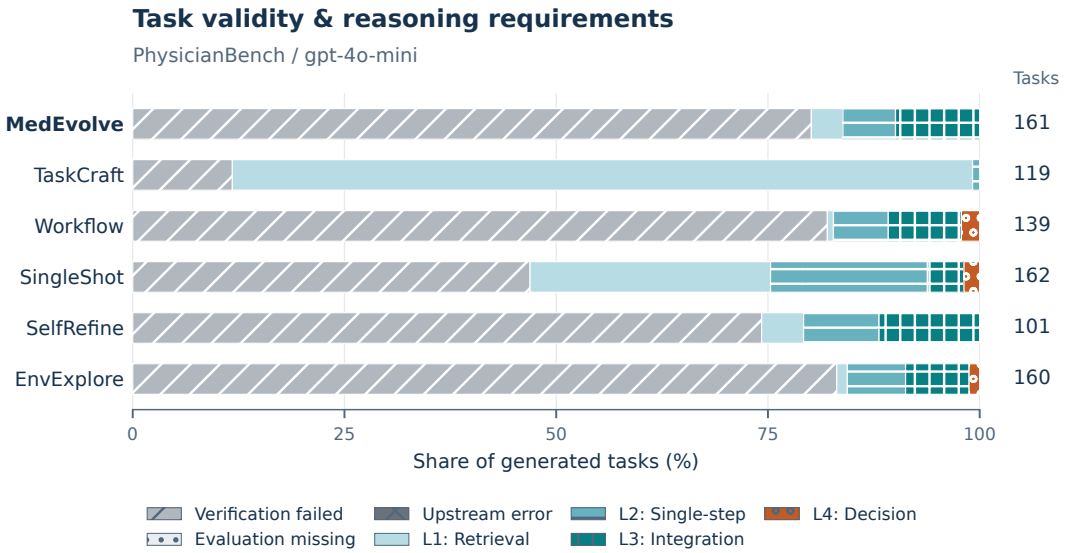


Figure 3: Verification status and reasoning depth on PhysicianBench with GPT-4o-mini (representative validity view).

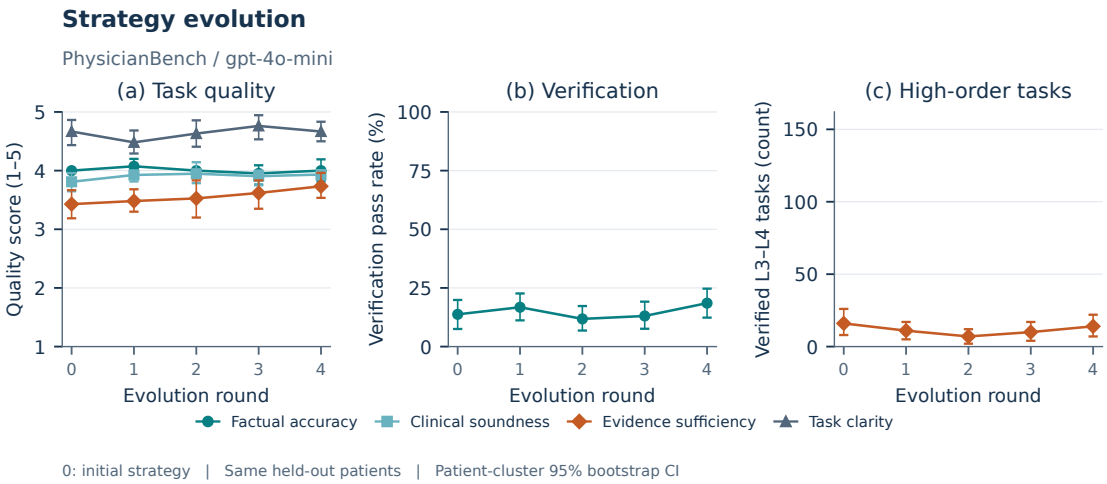


Figure 4: Descriptive held-out strategy-update curves (mixed effects): PhysicianBench, GPT-4o-mini.

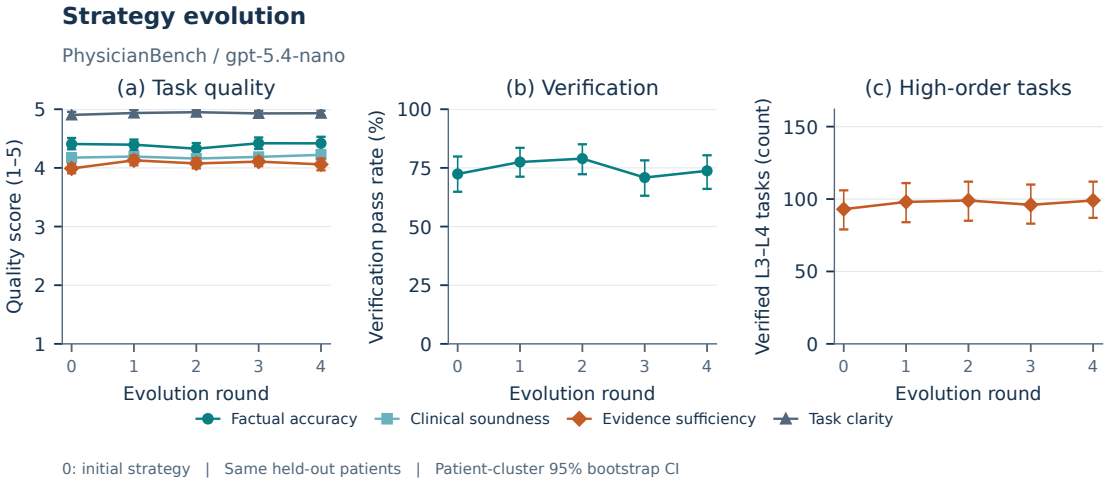


Figure 5: Descriptive held-out strategy-update curves (mixed effects): PhysicianBench, GPT-5.4-nano.

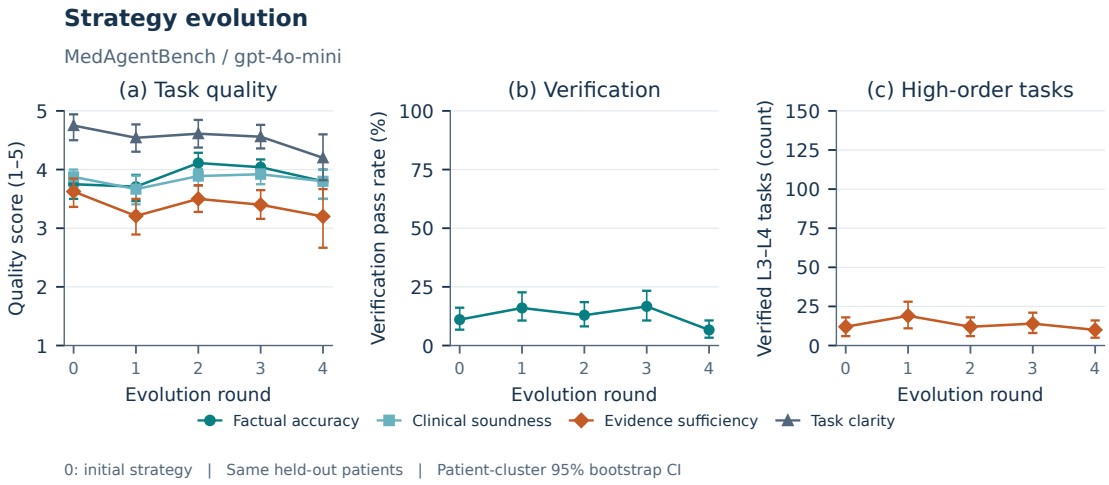


Figure 6: Descriptive held-out strategy-update curves (mixed effects): MedAgentBench, GPT-4o-mini.

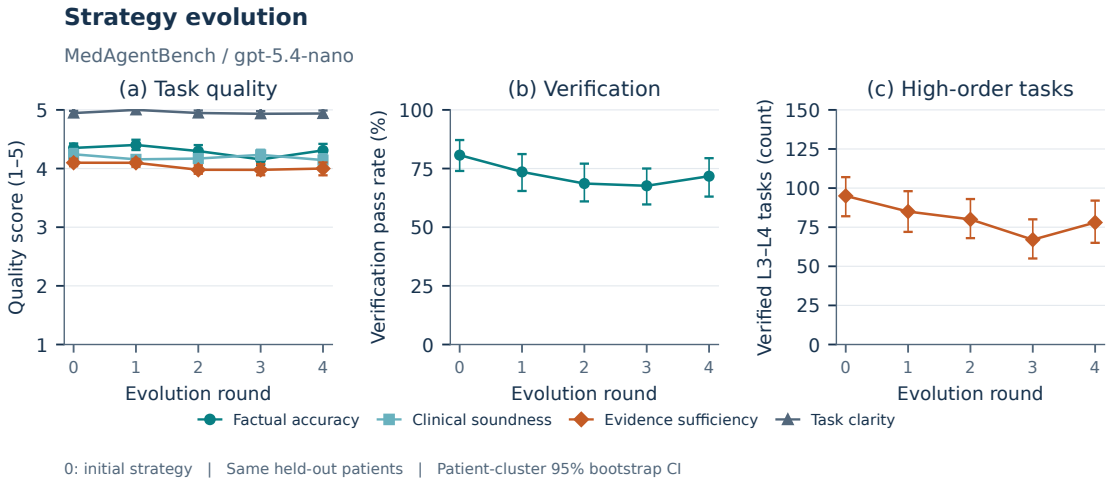


Figure 7: Descriptive held-out strategy-update curves (mixed effects): MedAgentBench, GPT-5.4-nano.